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Project Title: Ruby

Project Draft: Ruby as a Modern Programming Language

1. Introduction

This project shall discuss the Ruby programming language, its design philosophy, strengths, weaknesses and its relevance to software development today. The final paper must be written in the form of a tutorial where an introduction to the language as something unique and an addition to the analysis are to take place instead of merely restating information. In the last version, I intend to make the case that Ruby is still relevant due to its readability, productivity and good web development ecosystem despite its performance challenges and waning popularity when compared to some of the newer languages.

2. Background and History

In this part of the final report the origin of Ruby shall be explained in brief. Ruby is a programming language that was designed to enjoy programming among people or humans and was developed by Yukihiro Matsumoto in the 1990s. An important fact that I will discuss later is that Ruby is a language that integrates the concepts of object-oriented, procedural and scripting languages. As will be demonstrated in the final paper, the philosophy of developer happiness that Ruby adopted made it particularly effective in the web development with Ruby on Rails.

3. Main Features of Ruby

In the completed report, I will discuss Ruby's major language features:

Feature	Planned discussion
Pure object-oriented model	Everything in Ruby is treated as an object, which makes the language conceptually consistent.
Simple, readable syntax	Ruby is often praised for code that reads almost like English.
Dynamic typing	This increases flexibility and speed of development, but can also create maintenance challenges in larger systems.
Blocks, iterators, and metaprogramming	These features make Ruby expressive and powerful, especially for DSLs and web frameworks.

Short code examples will also be included in the final report to a similar section to show readability and flexibility.

4. Applications of Ruby

Large portion of the final paper will dwell on the where Ruby is used. Ruby is yet closely connected with web application development, particularly using Rails. It finds application in automation, scripting, DevOps, and rapid prototyping. According to recent studies, Ruby still is useful as its ecosystem allows carrying out practical developmental work, despite being no longer the hype in the industry (Akbarpour et al., 2025). In the final report, I will make a comparison of the practical strength of Ruby with other languages like Python and JavaScript.

5. Strengths of Ruby

I would state that the greatest strength of Ruby is its productivity as a developer. Ruby enables programmers to create an elegant and compact code which is helpful in startups, prototypes and maintainable web apps. Akbarpour (2025) reveals that most of the developer challenges in Ruby revolve around the need to comprehend, maintain, and fix existing applications, which makes it seem the language takes an active role to warrant further research. The developed Rails ecosystem is another strength that is capable of delivering a full-stack framework to create database driven applications efficiently (Takahashi, 2025).

6. Limitations and Criticism

In this part of the final report, the weaknesses of Ruby would be analysed. Performance is the most objectionable criticism. Ruby tends to be slower than compiled programming languages like C++ or Rust and can also be slower than certain high-performance environments. The second weakness is that the popularity of Ruby has declined compared to newer ecosystems that can influence the job demand and development of communities (Akbarpour et al., 2025). I will contend that reduced popularity will not imply that the language is no longer useful rather that Ruby can now have a more specialized though possibly useful position.

7. Planned Conclusion

The final report will draw the conclusion that Ruby is still a significant programming language due to its human centered design, expressive syntax, and powerful framework support. Nevertheless, the paper shall also admit that Ruby has to compete with more agile language and ecosystems. The last argument I will make is that Ruby is not to be interpreted as the language of the future which will be dominant. However, it will retain

its tangible value in those areas where programmer output and readability of code have the greatest importance.

8. References

Akbarpour, N. (2025). *From understanding to repair: An empirical and agentic investigation of Ruby developer challenges*. University of British Columbia.

Akbarpour, N., Saleem Mirza, A., Raoofian, E., Fard, F. and Rodríguez-Pérez, G., (2025), June. Unveiling ruby: Insights from stack overflow and developer survey. In *Proceedings of the 29th International Conference on Evaluation and Assessment in Software Engineering* (pp. 580-591).

Takahashi, K., (2025). A Comparative Case Study of Two Pedagogical Approaches in Web Development Education: From a Traditional Java Environment to a Modern Ruby on Rails Ecosystem. *Science, Engineering and Technology*, 5(2), pp.20-28.